CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF TECHNOLOGY & ENGINEERING

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Manufacturing Planning And Control (ME763)

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Manufacturing Planning and Control (ME763) **Date:**

Experiment No 1
Introduction to Minitab

AIM:
Introduction to Minitab

HISTORY:

Minitab is Statistical Analysis software that allows to easily conducting analyses of data. This is one of the suggested software for the class. This guide is intended to guide you through the basics of Minitab and help you get started with it.

Minitab is a statistics package developed at the Pennsylvania State University by researchers Barbara F. Ryan, Thomas A. Ryan, Jr., and Brian L. Joiner in 1972. It began as a light version of OMNITAB, a statistical analysis program by NIST; the documentation for OMNITAB was last published 1986, and there has been no significant development since then

Minitab is distributed by Minitab Inc, a privately owned company headquartered in State College, Pennsylvania. Minitab Inc. also produces Quality Trainer and Quality Companion, which can be used in conjunction with Minitab the first being an eLearning package that, teaches statistical tools and concepts in the context of quality improvement, while the second is a tool for managing Six Sigma and Lean Manufacturing.

REFERENCES:

1. Minitab, Inc. "MINITAB release 17: statistical software for windows." *Minitab Inc, USA* (2014).

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Experiment No 2
Graphing Data

AIM:
Graphing of data with Minitab

OBJECTIVES:
- Create, interpret, and edit histograms
- Create and interpret scatterplots with the Minitab Assistant
- Arrange multiple graphs on one page
- Save a project

EXPLORE THE DATA: (For Histogram)

1. Start Minitab.
2. Choose **Help > Sample Data.**
3. Double-click the Getting Started folder, and then double-click Shipping Data. MTW
4. If you have your own Data than Write into Worksheet.
5. Choose **Graph** > **Histogram.**
6. Click **with Fit**, and then click **OK.**
7. In **Graph variables**, select Temperature.
8. Click **Multiple Graphs.**
9. On the **By Variables** tab, in **by variables with groups in separate panels**, Select Ice Cream Sales.
10. Click **OK** in each dialog box.

(For Scatterplot)

1. Start Minitab.
2. Continue With Same Step Write your Data into Worksheet.
3. Choose **Graph** > **Scatterplots.**
4. Click **with Regression**, and then click **OK.**
5. In **Y variables**, select Temperature.
6. In **X variables**, Select Ice Cream Sales.
7. Click **OK** in each dialog box.

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QUESTIONS:

The local ice cream shop keeps track of how much ice cream they sell versus the noon temperature on that day. Here are their figures for the last 12 days:

Ice Cream Sales vs. Temperature	Temperature °C	14.2	16.4	11.9	15.2	18.5	22.1	19.4	25.1	23.4	18.1	22.6	17.2
	Ice Cream Sales	215	325	185	332	406	522	412	614	544	421	445	408

REFERENCES:

1. Minitab, Inc. "MINITAB release 17: statistical software for windows." *Minitab Inc, USA* (2014).

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Experiment No 3
Analyzing Data

AIM:
Analyzing data with Minitab

OBJECTIVE:
- Summarize the data
- Compare means
- Access Stat Guide
- Use the Project Manager

SUMMARIZE THE DATA: (For Basic Statistics)

1. Start Minitab.
2. Choose **Help > Sample Data**.
3. Double-click the Getting Started folder, and then double-click Shipping Data. MTW
4. If you have your own Data than Write into Worksheet.
5. Choose **Stat > Basic Statistics > Display Descriptive Statistics.**
6. In **Variables**, Select NBA teams.
7. Click **Statistics.**
8. Tick Mark as per Finding Requirements, **Median** and **Mode.**
9. Click **Ok** in each dialog box.

(For ANOVA):

1. Start Minitab.
2. Write Data into Worksheet. (Derive in two or more column)
3. Choose **Stat > ANOVA > One-Way.**
4. In **Response** and **Factor**, Select NBA teams.
5. Click **OK** in each dialog box.

QUESTIONS:

The following are the top 20 player salaries (in millions of dollars) for 2014 - 15 NBA teams, according to ESPN. They are ordered from largest to smallest.

NBA team	23.5	23.4	23.2	22.5	21.4	20.6	20.6	20.1	19.8	19.3
	19	18.9	17.7	16.5	16	15.9	15.8	15.7	15.7	15.7

(a) Using only the data above, what is the median salary of the top 20 player salaries?
(b) What is the mode salary of the top 20 player salaries?

REFERENCES:

1. http://espn.go.com/nba/salaries

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Experiment No 4
Assessing Quality

AIM:
Assessing Quality with Minitab

OBJECTIVES:

- Create and interpret control charts
- Add stages to a control chart
- Update a control chart
- Add date/time labels to a control chart
- Perform and interpret a capability analysis

CREATE AN XBAR-S CHART:

1. Start Minitab.
2. Write your Data into Worksheet.
4. Choose **Stat > Control Charts > Variables Charts for Subgroups > Xbar-S**.
5. From the drop-down list, select **all observations for a chart are in one column**, then select Temperature.
6. In **Subgroup sizes**, Select Red Bull Sales or 10.
7. Click **OK**.

QUESTIONS:

The Famous Red Bull shop keeps track of how much Tin they sell versus the evening temperature on that day. Here are their figures for the last 12 days:

Red Bull Sales vs Temperature	Temperature °C	14.2	16.4	11.9	15.2	18.5	22.1	19.4	25.1	23.4	18.1	22.6	17.2
	Red Bull Sales	215	325	185	332	406	522	412	614	544	421	445	408

REFERENCES:

1. Minitab, Inc. "MINITAB release 17: statistical software for windows." *Minitab Inc, USA* (2014).

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Experiment No 5
Designing an Experiment

AIM:
Study of Designing an Experiment

OBJECTIVES:
- Learn about designed experiments in Minitab
- Create a factorial design
- View a design and enter data in the worksheet
- Analyze a design and interpret the results
- Use a stored model to create factorial plots and predict a response

CREATE AND SELECT A DESIGNED EXPERIMENT:

1. Choose **File > New > Project**.
2. Choose **Stat > DOE > Factorial > Create Factorial Design**.
3. Click **Display Available Designs**.
4. Click **OK** to return to the main dialog box.
5. Under **Type of Design**, select **2-level factorial (default generators).**
6. From **Number of factors**, select **2**.
7. Click **Designs**.
8. From **Number of replicates for corner points**, selecting **3**.
9. Click **OK** to return to the main dialog box.
(Now all buttons are enabled)
10. Click **Factors**.
11. In the row for **Factor A**, under **Name**, enter OrderSystem. Under **Type**, select **Text**. Under **Low**, enter New. Under **High**, enter Current.
12. In the row for **Factor B**, under **Name**, enter Pack. Under **Type**, select **Text**. Under **Low**, enter A. Under **High**, enter B.
13. Click **OK** to return to the main dialog box.
14. Click **Options**.
15. In **Base for random data generator**, enter 9.
16. Verify that **Store design in worksheet** is selected.
17. Click **OK** in each dialog box.

18. In the worksheet, click the column name cell of C7 and enter Hours.
19. In the Hours column, enter the data as shown below.
20. Choose **Stat > DOE > Factorial > Analyze Factorial Design.**
21. In **Responses**, enter Hours.
22. Click **Terms**. Verify that **A:OrderSystem, B:Pack**, and **AB** are in the **Selected Terms** box.
23. Click **OK**.
24. Click **Graphs**.
25. Under **Effects Plots**, select **Pareto** and **Normal**.
26. Click **OK** in each dialog box.

QUESTIONS:

Hours	10	12	08	22	24	21	18

REFERENCES:

1. Minitab, Inc. "MINITAB release 17: statistical software for windows." *Minitab Inc, USA* (2014).

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Experiment No 6
Using Session Commands

AIM:
Study of the Session Command

OBJECTIVES:
- Enable and enter session commands
- Perform an analysis using session commands
- Re-execute a series of session commands with the Command Line Editor
- Create and run an exec file

ENABLE SESSION COMMAND:

1. Start Minitab.
2. Choose **Help > Sample Data.**
3. Double-click the Getting Started folder, and then double-click Session Commands. MTW
4. Click the Session window to make it active.
5. Choose **Editor > Enable Commands.**
6. (Optional) Enable Session commands by default for all Minitab sessions.

a. Choose **Tools > Options**. Expand **Session Window**, and then select **Submitting Commands**.
b. Under **Command Language**, click **Enable**.

QUESTIONS:

The local ice cream shop keeps track of how much ice cream they sell versus the noon temperature on that day. Here are their figures for the last 12 days:

Ice Cream Sales vs Temperature	Temperature °C	14.2	16.4	11.9	15.2	18.5	22.1	19.4	25.1	23.4	18.1	22.6	17.2
	Ice Cream Sales	215	325	185	332	406	522	412	614	544	421	445	408

REFERENCES:

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Experiment No 7
Generating a Report

AIM:
Generate a Report in Minitab

OBJECTIVES:
- Add a graph to the Report Pad
- Add Session window output to the Report Pad
- Edit a report
- Save a report
- Copy the Report Pad contents to a word processor
- Send output to Microsoft PowerPoint

STEP FOR GRAPH TO THE REPORTPAD:

1. Choose **File > Open**.
2. Browse to C:\Program Files (x86)\Minitab\Minitab 17\English\Sample Data\Getting Started. (Adjust this file path if you chose to install Minitab to a location other than the default.
3. Double-click Reports.MPJ.
4. Choose **Window > Histogram of Days**.
5. Right-click the graph, then choose **Append Graph to Report**.
6. Choose **Window > Project Manager**.
8. Click the **ReportPad** folder. The histogram is added to the ReportPad.
9. Choose Window > Session.
10. In the Session window, click in the output for Results for Center = Central, right-click, then choose Append Section to Report. Sections of Session window output are separated by titles, which are in bold text.
11. Repeat the steps above for Results for Center = Eastern and Results for Center = Western.
12. Choose Window > Project Manager, then click the ReportPad folder. Maximize the window to see more of your report.

REFERENCES:

1. Minitab, Inc. "MINITAB release 17: statistical software for windows." *Minitab Inc, USA* (2014).

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Experiment No 8
Preparing a Worksheet

AIM:
Prepare a worksheet in Minitab.

OBJECTIVES:
- Open a worksheet
- Open an Excel file
- Open a text file
- Combine the data into one worksheet
- Move and rename a column
- Recode the data
- Insert and name a new data column
- Assign a formula to a column

STEP FOR PREPARING WORKSHEET INTO DIFFERENT OBJECTS:

1. Start Minitab.
2. Choose **Help > Sample Data**.
3. Double-click the Getting Started folder, and then double-click Eastern.MTW.
4. Choose **File > Open**
5. Double-click Central.xlsx
6. Click **OK**.
7. Choose **File > Open**.
8. Double-click Western.txt
9. Click **OK**.
10. Choose **Data > Stack Worksheets**.
11. From Stack option, select Stack worksheets in a new worksheet.
12. Use the arrow buttons to move the three worksheets From Available Worksheets to Worksheets to stack.
13. In New worksheet name, enter MyShippingData.
14. Click **OK**.
15. Click in the Source column, then choose **Editor > Move Columns**.
16. Under Move Selected Columns, select before column C1.

17. Click **OK**.
18. Click in the column name cell Source, type Center, and press Enter.
19. Choose **Data > Recode > To Text**.
20. In Recode values in the following columns, enter Center.
21. From Method, select Recode individual values.
22. Under Recoded Value, replace Eastern.MTW with Eastern.
23. Under Recoded value, replace Western.txt with Western.
24. From Storage location for the recoded columns, select In the original columns.
25. Click **OK**.
26. Click any cell in C4 to make that column active.
27. Right-click, then choose Insert Columns.
28. Click in the name cell of C4. Type Days, then press Enter.
29. Choose **Calc > Calculator**.
30. In Store result in variable, enter Days.
31. In Expression, enter Arrival – Order.
32. Select Assign as a formula.
33. Click OK.
34. Click in the worksheet, then choose **File > Save Worksheet** As Browse to the folder that you want to save your files in.
35. In File name, enter MyShippingData.
36. From Save as type, select Minitab.
37. Click **Save**.

REFERENCES:

1. Minitab, Inc. "MINITAB release 17: statistical software for windows." *Minitab Inc, USA* (2014).

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Experiment No 9
Customizing Minitab

AIM:
Customizing Minitab.

OBJECTIVE:

- Change default Options for graphs
- Create a custom toolbar
- Add commands to a custom toolbar
- Assign shortcut keys for a menu Command
- Restore Minitab's default options

ADDING AUTOMATIC FOOTNOTE:

1. Start Minitab.
2. Choose **Tools > Options**. Expand **Graphics**, expand **Annotation**, then select **My Footnote**.
3. Under **Information to include in my footnote**, select **Worksheet name** and **Date the graph was last modified**.
4. In **Custom text**, enter Shipping center efficiency.
5. Click **OK**.

(Create a histogram to view the footnote)
6. Choose **Graph > Histogram.**
7. Click **With Fit**, then click **OK**.
8. In **Graph variables**, enter Days.
9. Click **Multiple Graphs**.
10. On the **By Variables** tab, in **By variables with groups in separate panels**, enter Center.
11. Click **OK** in each dialog box.

(Create a toolbar)
12. Choose **Tools > Customize.**
13. On the **Toolbars** tab, click **New**.
14. In **Toolbar Name**, enter Shipping Data.
15. Click **OK.**

(Add commands to the toolbar)
16. Drag the blank toolbar to dock it be side an existing Minitab toolbar.
17. On the **Commands tab**, under **Categories**, select **Graph**.
18. Under **Commands**, select **Histogram**.
19. Drag **Histogram** to the new toolbar.
20. Under **Categories**, select **Assistant**.
21. Under **Commands**, select **Scatterplot (Groups)**.
22. Drag **Scatterplot (Groups)** to the new toolbar.
23. Click **Close**.

(Assign a shortcut key)
24. Choose **Tools > Customize.**
25. On the Keyboard tab, from **Category**, select **Graph**.
26. Under **Commands**, select **Histogram**.
27. Click in **Press New Shortcut Key**.
28. Press **Ctrl+Shift+H**
29. Click **Assign**. (The new shortcut key appears under Current Keys.)
30. Click **Close**.

(Restore Minitab's default options)
31. Choose **Tools > Manage Profiles**.
32. Move My Profile from **Active profiles** To **Available profiles.**
33. In **Available profiles**, double-click My Profile and enter Shipping Center Analysis.
34. Click **OK.**

REFERENCES:

1. Minitab, Inc. "MINITAB release 17: statistical software for windows." *Minitab Inc, USA* (2014).

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